The Lexifyd Polysemy Challenge

The Problem
In Tamil, a single word often carries vastly different meanings based on its context (**Polysemy**).

- **படி** (padi): Can mean *read*, *staircase*, *a measure*, or *settle/ deposit*.
- **திங்கள்** (thingal): Can mean *Monday*, *the moon*, or *month*.
- **ஆறு** (aaRu): Can mean *six*, *a river*, or the act of *cooling down/ comforting*.

For example, the word 'ஆறு' can mean the number *six*, a *river*, or the act of *cooling down/ comforting*. Current word-based games, rely on static "Flashcard" models (Word A = Meaning B), which fail to teach learners how to navigate these real-world linguistic nuances.

The Mission
Design and implement a **Context-Aware Lexical Game Engine** for *Lexifyd*. Your goal is to move from generating "Static Pairs" to "Dynamic Sentences", which can be used as game to test user's understanding of the different meanings.

The Task:
- **The Data Engine:** Build a pipeline that takes an ambiguous Tamil word as input and uses NLP/LLMs to generate:
    1. Two or more distinct **Sentences** where the word is used in different senses.
    2. **Smart Distractors:** Generate "wrong" options that are grammatically correct in Tamil but semantically nonsensical in that specific sentence.
- **The Mini-Game UI:** A functional prototype where a learner must drag the correct "Meaning" into the correct "Sentence Slot."

Bonus Task: The "Semantic Web" Visualization
Don't just show a list. Create an innovative **Visualization Model** (e.g., a Knowledge Graph, a "Meaning Star," or a 3D Cluster) that allows a user to explore how one root word branches out into different meanings along with some context of these meanings.

Technical Constraints & Expectations
- **Automation is Key:** Solutions that use a "hard-coded" database will score lower than those using a generative or algorithmic approach (e.g., using OpenAI API, Google Gemini API, or Tamil-specialized NLP libraries like *IndicNLP*).
- **WordNet:** You may use Tamil WordNet as a direct resource or through API as required in addition to some LLM engine
- **Morphological & POS Awareness:** Tamil words change suffixes based on tense and gender (e.g., படித்தான் vs படிப்பார்கள் vs படிக்கிறாள்). Your engine should ideally handle these variations and be able to generate examples in these different variations to test the learner. In addition, it is also important to identify the word meanings across different POS categories (a *noun* and a *verb*).
- **The "Human-in-the-Loop" Metric:** How does your system ensure the generated sentences aren't "hallucinations"?

Dataset Sample

Here is a sample dataset with input and expected outputs.

Input Word	Identified Context (Sense)	Generated Challenge
படி (Padi)	Verb: *to read*	அவன் தேர்வுக்காகப் பாடம் _____. Options: படிக்கிறான், ஏறினார், அளக்கிறாள், படிந்தது
படி (Padi)	Noun: *step/ stair*	கோவில் _____ உயரமாக இருந்தது. Options: படியில், படி, படிப்பில், படிப்பு
கல் (Kal)	Noun: *stone*	சிற்பி _____ சிலையாக செதுக்கினார். Options: கல்லை, கற்பதை, கல்விக்கு, கல்லில்
கல் (Kal)	Verb: *to learn*	நல்ல பழக்கங்களை நாம் _____ வேண்டும். Options: கற்க, கல்லை, விற்க, நிற்க
ஆறு (Aaru)	Noun: *a river*	மீனவர் _____ மீன் பிடித்தார். Options: ஆற்றில், அடியில், ஆறுக்கு, ஆறாக

The task is to:

- Identify Two or more contexts/ senses for the given ambiguous word (possibly in different POS categories)
- Generate a challenge question to be used in the game feature to test the user understanding. The question should include the correct answer along with 3 intelligently designed distractors (incorrect answers)
- Use the above challenge questions in a prototype game UI, to test the users. Track user performance and provide resonse summary to the user as feedback.

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"hackathon_title": "Lexifyd Contextual Intelligence Challenge",

"data_description": "A seed list of ambiguous Tamil words for context-aware game generation.",

"seed_words": [

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